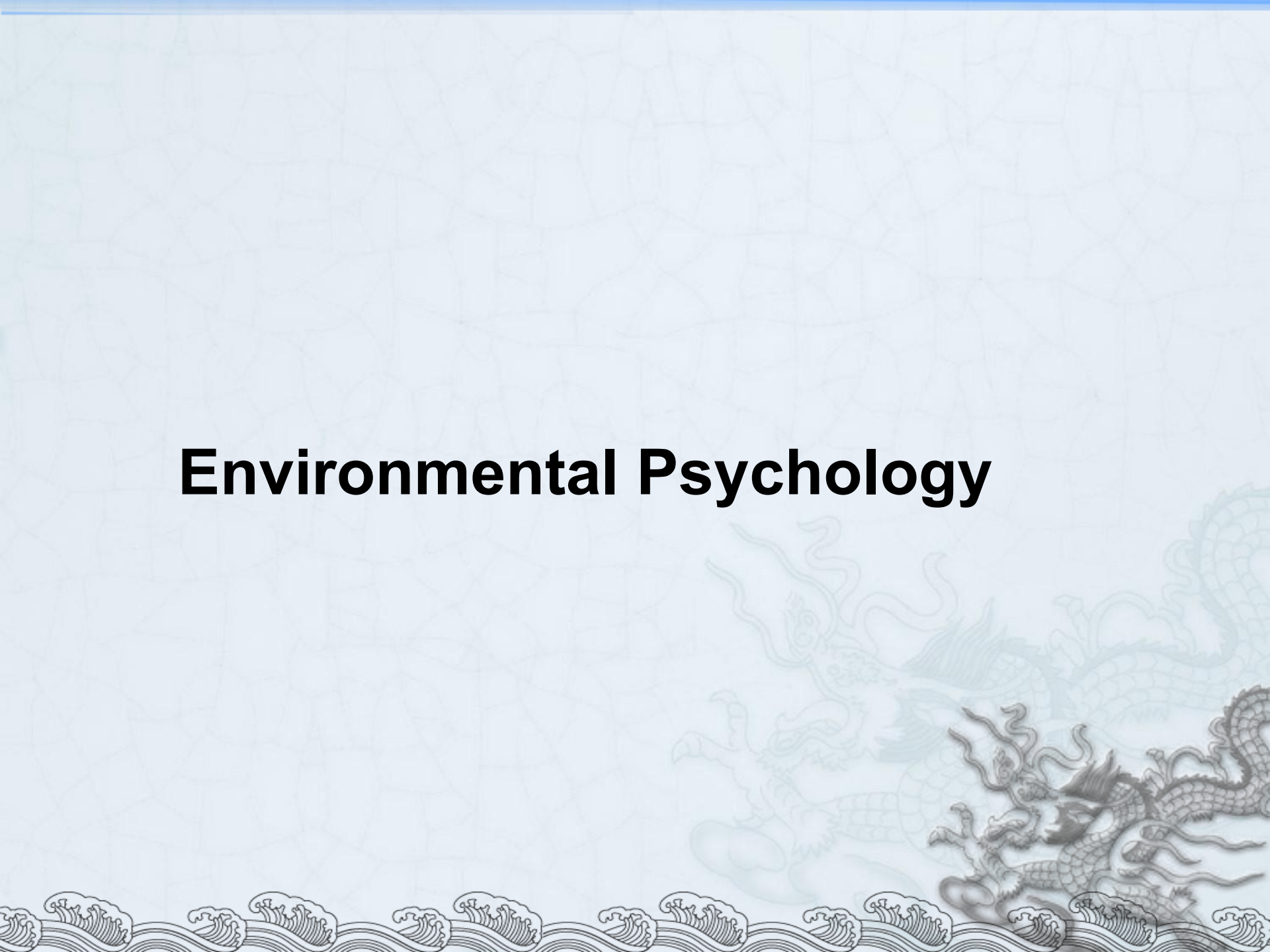
Environmental Psychology



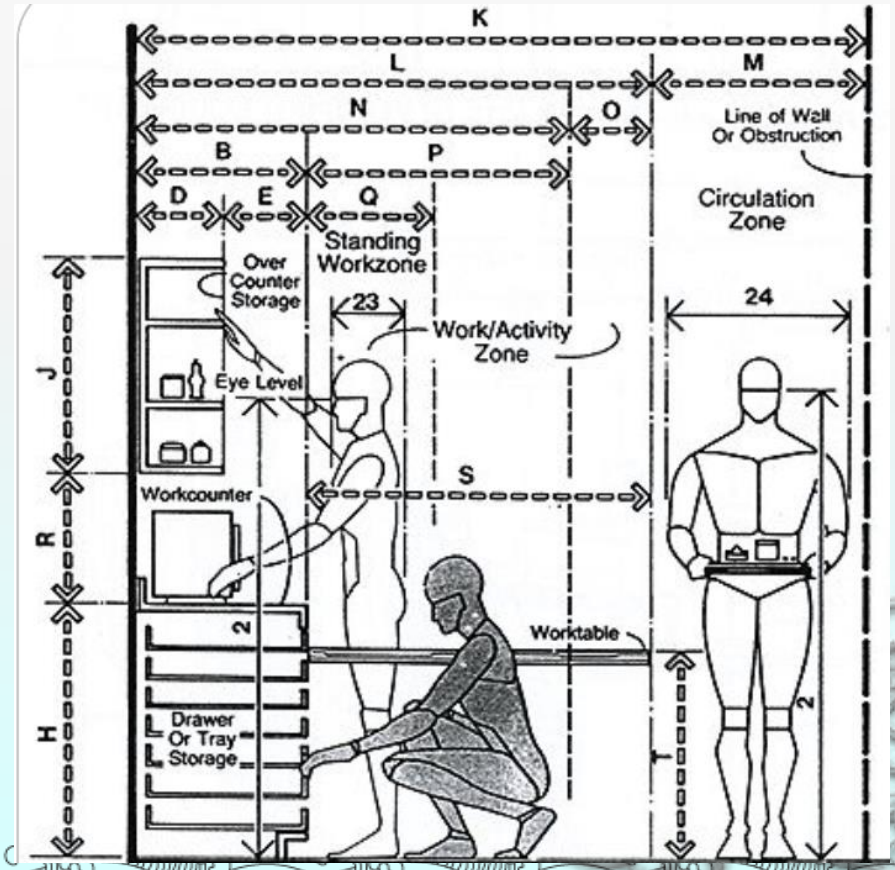
Lecture six

- Personal space
- Territoriality
- Density & Crowding
- Environmental Stress



Personal space - definition

- The term “personal space” was coined by Katz， which is rooted in biology.
- An overload interpretation of why we maintain an interpersonal distance between ourselves and other is that such spacing is necessary to avoid overstimulation.
- According to this notion, too close a proximity to others causes us to be bombarded with excessive social or physical stimuli.



- The distance between individuals determines the quality and quantity of stimulation that is exchanged.
- Distance communicates information about the type of relationship between individuals.

Personal space - definition

- The stress approach assumes that we maintain personal space to avoid a variety of stressors associated with too close a proximity.
- The arousal conceptualization posit that when personal space is inadequate, individuals experience too much arousal.
- The behaviour constraint approach suggests that personal space is maintained to prevent our behavioural freedom from being impinged upon.

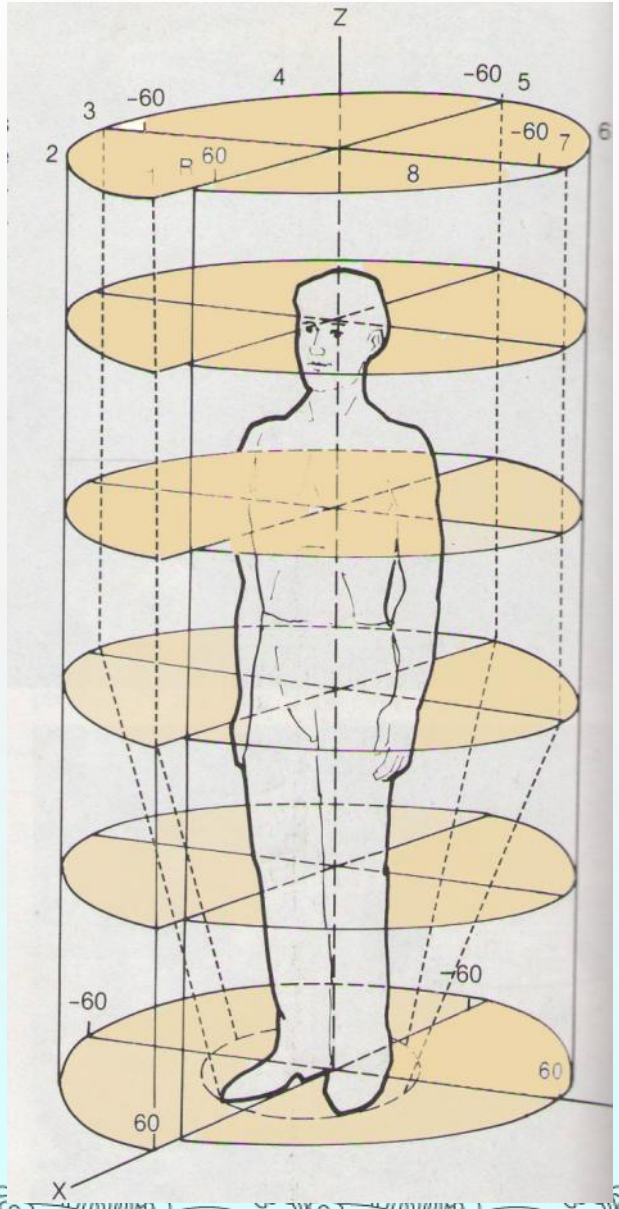








Determinant of personal space



- A three-dimensional view of the shape of **personal space** - analogous to a bubble fulfilling a number of functions.
- It is generally cylindrical, and of a consistent shape above our waist, but seems to taper off below it.

- Varying spatial zones and the amount of space we desire between ourselves and others expands and contracts depending on the situation.
- It is not the case that personal space is the same size for all individuals and in all situations. *For instance, a distance between people and desks.*

- **Personal space** also must focus on other behaviors. *Except concerning with distance, studying personal space also must focus on other behaviors, such as body orientation and eye contact, in order to get a complete understanding of our spatial behavior.*

Territoriality

- One way of viewing territories is as places which are owned or controlled by one or more individuals. **Human territoriality can be viewed as a set of behaviors and cognitions an organism or group exhibits**, based on perceived ownership of physical space, as used here may refer either to actual ownership or to control over space.
- Territorial behaviors serve important motives and needs for the organism and include occupying an area, establishing controlling over it, and in some cases defending it.
- Personal space tends to be invisible, movable, body-centered, and regulates how closely individuals will interact. Territory is visible, stationary, and tends to be home-centered regulating who will interact.
- Territorial behavior is practiced by humans and animals. Some researchers consider human territoriality to be *instinctive*, as it is in animals; some consider it to be *learned*, and some consider it an *interaction* of the two.
- Both species have a drive to claim and defend territory.

Three Types of Territoriality

- There are three types of territories used by humans. Primary territories are most important, followed by secondary and public territories.
- They differ in the duration of occupancy, the cognitions they foster in the duration of occupancy, the cognitions they foster in the occupant and others, the amount of personalization, and the likelihood of defense if violated.

	Extent to which territory is occupied /extent of perceived ownership by self and others	Amount of personalization/ likelihood of defense if violated
Primary Territory (e.g. home, office)	High. Perceived to be owned in a relatively permanent manner by occupant and others	Extensively personalized; owner has complete control and intrusion is a serious matter
Secondary Territory (e.g., classroom)	Moderate. Not owned; occupant perceived by others as one of a number of qualified users	May be personalized to some extent during period of legitimate occupancy; some regulatory power when individual is legitimate occupant
Public Territory (e.g., area of beach)	Low. Not owned; control is very difficult to assert, and occupant is perceived by others as one of a large number of possible users	Sometimes personalized in a temporary way; little likelihood of defense

Functions of Territoriality *for humans*

- Humans are more flexible with respect to use of territories for functions.

For People in …..	Organizing Function of Territory
Public places (e.g., a library, the beach)	Organizes space; provides an interpersonal distancing mechanism
Primary territories (e.g., a bedroom)	Organizes space by providing a place which promotes solitude; allows intimacy; expresses personal identity
Small face-to-face groups (e.g., the family)	Clarifies the social ecology of the group and facilitates group functioning; may provide home court advantage
Neighborhoods and communities	Promotes an "in-group" who "belong" and can be trusted; differentiates them from an "out-group" who don't belong and can't be trusted. In some urban areas, territorial control makes a space safe to use



House: primary and small face to face group


RONALDINHO
10
in-group



Small face to face group in-one building (Unitè d'Habitation in Marseille)

Functions of Territoriality *for animals*

- Animals maintain territory for important functions such as mating, dispersing the population more evenly, food gathering and protecting food supplies, shelter, rearing of young, and controlling intra-species aggression. Territories are often quite essential to survival for them, and animals tend to defend territories vigorously when violations occur.



Effects of population density

- For animals, high density is generally aversive.



food



Rearing young



Water supply

- For humans, it depends more on the situations. For the most part, the effects of high densities on people are neither severe nor uniform.

Effects of population density on humans

- *Affect:* It was observed that subjects reported more anxiety in the dense than in the non-dense conditions, although this probably doesn't surprise anyone who has ever had to perform a task with hordes of others "breathing down one's neck."
- Women may approach crowded settings in more cooperative ways than do men.
- **Crowding versus Density***:* Density is an objective measure of individuals per unit of area. It can be measured at different physical scales. The room, building, neighborhood, city, region, and nation in which an individual is located probably all have different densities. Concerned on human behavior. **Crowding** refers to a person's experience of the number of other people around.





Crowding

- **Crowding** meant the formation of large, temporary groups of emotional individuals. These could include swarms of peaceful concert-goers or fair patrons, but more often, **they meant lynch mobs, rioters, or panic-stricken disaster victims**.
- *Physiological arousal:* High density conditions, subjects showed higher pulse rate and blood pressure readings than in more spacious conditions.

Don't people simply feel crowded when too many other people are around?

- **Influences on crowding:** many factors lead some individuals to feel crowded and others to feel uncrowded, even in the same objective setting
- Personal Influence - Personal control is an important component of crowding
- Another personality variable relevant to the crowding experience is affiliative tendency or sociability. Individuals who generally like to be with others seem to have a higher tolerance for dense situations than individuals who are less affiliative
- Psychiatric Status: autism

Crowding









Crowding Influences

- **Personal Influence**. *Preferences, Expectations, and Norms*
  - Crowding partly depends on how many people one expects or considers normal in a given situation. A person expects throngs of people at a fair, and thus does not feel crowded among many people there. However, that same person outdoors enjoying nature may feel crowded after encountering only a few other people.
  - Those who *preferred* higher densities felt less crowded; those who *expected* higher densities than they found felt less crowded.
  - *Experience.* Personal experience with high-density situations or familiarity with a behavior setting where crowding occurs may affect the degree of distress that is experienced. The trouble is, these effects are not always the same – sometimes experience helps and sometimes it does not.
  - *Gender, Mood, Culture*

- **Social Influence.** Individual's personal characteristics predispose them to experience more or less crowding, but the influence of others in that setting can either worsen or ameliorate crowding stress. Some of these social factors include:
  - The mere presence and behavior of nearby others;
  - Coalitions that tend to form in small groups (isolation);
  - The quality or type of relationship among the individuals receive (interpersonal similarity).

Satisfaction with roommate under crowded and uncrowded conditions

	Uncrowded	Crowded
Satisfaction with roommate	4.9	3.7
Perceived cooperativeness of roommate	4.7	3.9

* Higher numbers indicate more positive responses

Ratings of perceived similarity under crowded and uncrowded conditions

Sex	Crowded (high spatial density)	Uncrowded (low spatial density)
Male	5.7	4.4
Female	4.2	5.7

* Low numbers indicate greater perceived similarity

- **Physical Influence.** Another major influence on how crowded we feel is the physical setting. High density itself is the most obvious such factor. Other physical factors associated with crowding include:

  - scale is under investigation (room, building, neighborhood, city)

  - Architectural variations (ceiling height, furniture, arrangement, window placement, partitions, barriers, etc.)

  - Place variations (home, work, beach, etc.)

  - Weather;

  - Lighting;

  - Sound.



- **Privacy** means different things to different people. Privacy, as an ability to withdraw or separate ourselves for other people, is the desire to be able to be alone, having space that is away from where other people congregate; and having ability to **govern intimacy** and maintain **interpersonal distance**. More related to availability of **personal information** were the abilities to **keep secrets, limit self-disclosure**, and regulate when and with whom one might interact. (***personal space***)

- **Design features** which do not allow **sufficient privacy** may be associated with other difficulties. (e.g., fighting between crew members of the US aircraft carrier Kitty Hawk, crew quarters were not planned to allow for sufficient privacy and to meet other social needs).

- The **lack of privacy in prison** may **cause prisoners to learn antisocial behaviors**. Young prisoners may learn criminal behavior from older ones as a function of being housed in more open, **less private dormitory units**.


Meadow Hall,Sheffield
PIZZA
ENGLISH
fresh
Pizza &
usual


21
BERRY'S
21
19
BERRY'S
GEORG JENSEN
CHANEL
Chopard
MONT BLANC
BROSGILL OPTICIANS






PEDESTRIAN ZONE
No vehicles
Except for loading by
between 7am - 12 noon Mon-Sat
At any time
GERRARD'S CORNER
Cash Machine Inside
Cash Machine Inside

Environmental stress

- ***Stress*** as a process by which environmental events that threaten or challenge an organism's personal space (well-being) evoke various responses from the organism, as well as coping behavior directed toward the threat.

- The degree to which people will view an event as stressful is determined by appraisal.

- Physiological changes are part of environmental stress response, most reflecting increased arousal.

- Emotional, psychological, and behavioral changes may also occur as part of the stress response.

- *Cataclysmic events; personal stressors; Background stressors.*

- **Four important examples of environmental stress: *natural disaster, technological disaster, noise, and commuting to work.***



2014 Shanghai New Year's Eve stampede

（panic-stricken disaster）

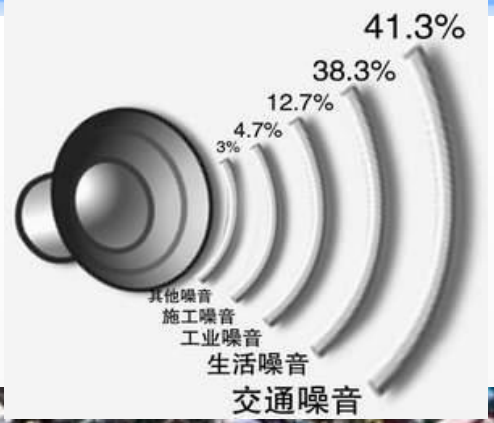
Natural disaster









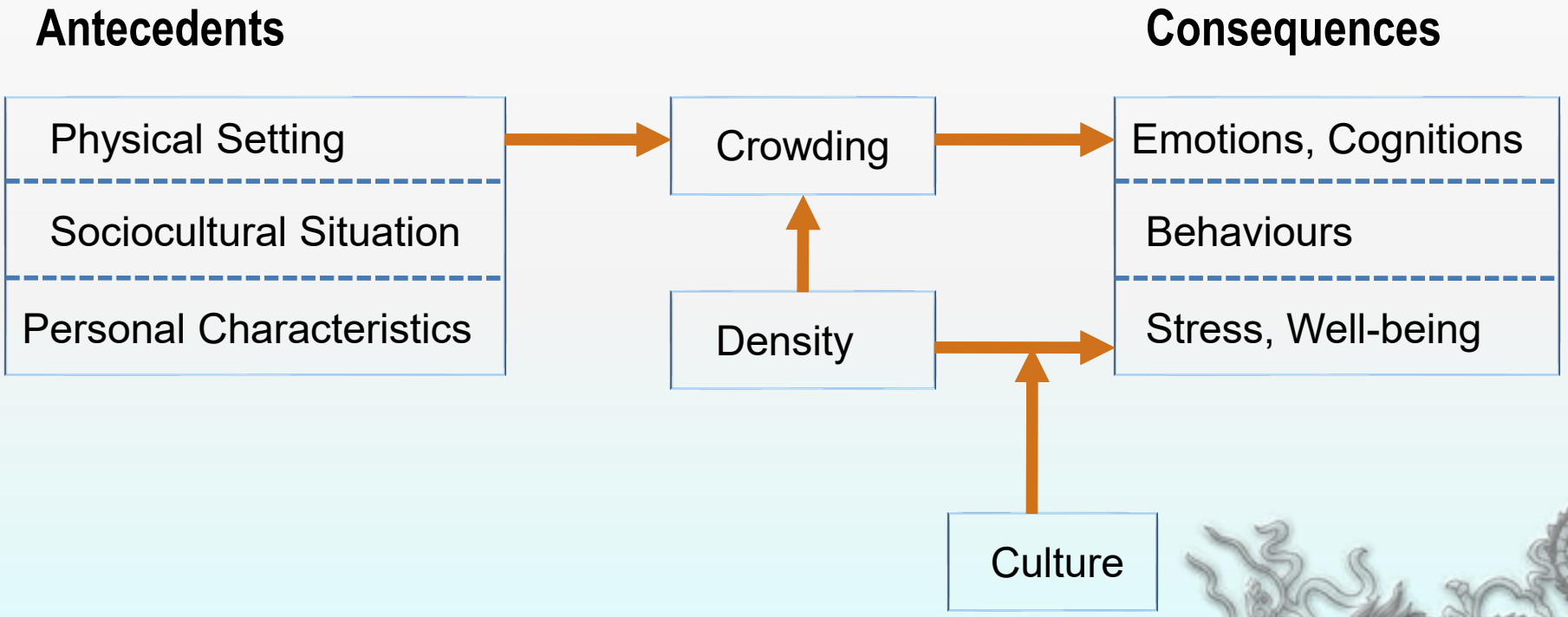




Commuting



Influence factors on crowding effects



STRESS VS ANXIETY

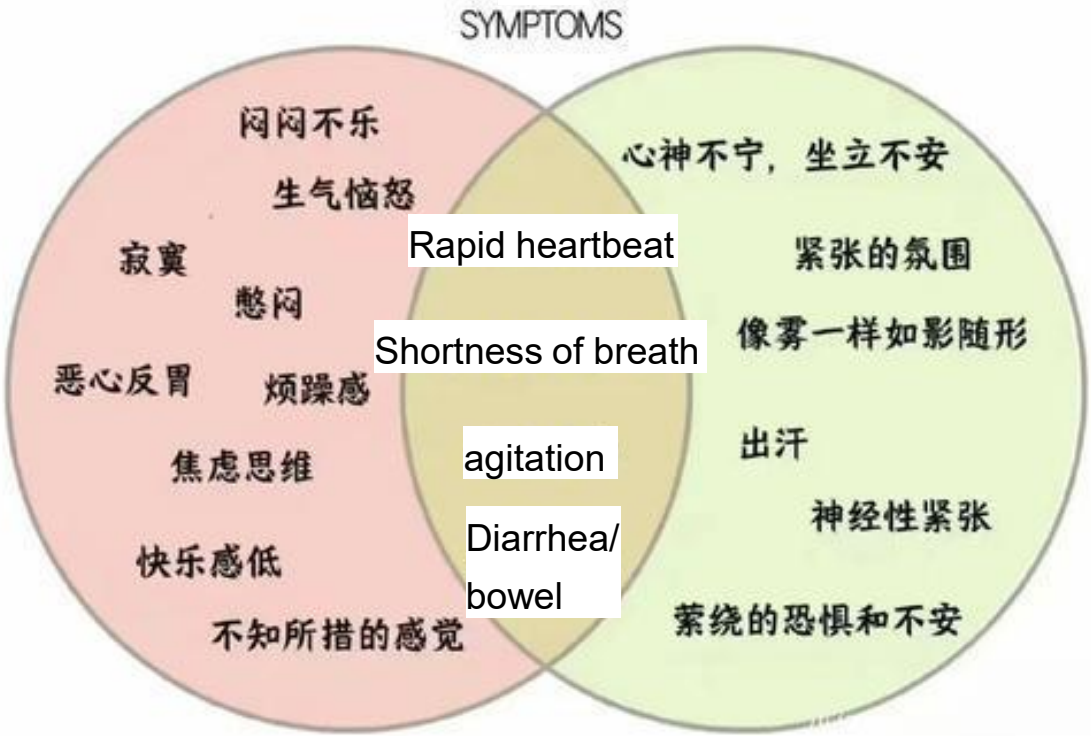
压力 焦虑

跨度

Short-term: tolerant it, adaptation → Long-term: sick, depression

对公认的潜在威胁作出反应

没有特定诱因或可识别的触发因素



Low mood;
Irritability/angry;
Loneliness;
Stifled;
Nausea;
Restlessness;
Anxious;
Feeling;
overwhelmed.

Unease;
Sense of tension;
pervasive;
Sweating;
Nervous;
Tension;
Lingering fear.

End of the Sixth Lecture

Thanks for attention

Any questions? (leilayu@hit.edu.cn)